Speech, Language & Dysarthria Clinical Assessment Summary

Report ID: SC-ASSESS-000004 | Assessment Date: October 27, 2026 at 04:24 UTC | Patient ID: SC-PT-0004

Patient Identifier:	SpeechCare Administrator (SC-PT-0004)	**Age & Biological Gender:**	None yrs / Female
Neurological History:	None reported	**Symptom Onset / Duration:**	Not specified
Prior Assessments:	None recorded	**Therapy / Treatment History:**	None recorded
Language / Dialect:	English	**Communication Needs:**	None noted

1. Audio Quality & Recording Validation

Quality Status:	ACCEPTABLE	**Signal-to-Noise Ratio:**	50.0 dB	**Clipping Check:**	Clear (No clipping)
Evaluated Task:	Sustained Vowel	**Speech Duration:**	8.5s total (2.6s active speech)		

2. Dysarthria Severity Estimation & Speech Production Analysis

ESTIMATED SEVERITY: NO CLEAR DYSARTHRIC INDICATORS DETECTED **Calculated Confidence:** 1.0%

Intelligibility Index:	44.1% (Estimated)	**Speech Rate:**	87.5 WPM (2.19 syl/sec)
Articulation Accuracy:	40.6/100 (Severe Consonant Blu)	**Vocal Quality:**	Moderate Dysphonia / Vocal Tremo
Prosody / Intonation:	Natural Dynamic Intonation	**Respiration Coordination:**	Shortened Phrasing with Frequent

Contributing Features: Severely reduced intelligibility index (44.1%).; Profoundly slowed speech rate (87.5

Notice: Severity estimate reflects acoustic screening metrics and is NOT a definitive clinical diagnosis.

3. Screening for Associated Speech & Language Disorders (Differential Profile)

Condition Screened	Screening Index	Clinical Distinction	Evaluation Status
Apraxia of Speech (AOS):	70.0%	Motor Planning vs Execution	Screening Indicator (Not a Diagnosis)
Aphasia (Language):	87.0%	Linguistic Formulation / Retrieval	Requires Bedside Language Battery
Stuttering / Disfluency:	74.0%	Speech Fluency Breaks & Blocks	Disfluency Frequency Analysis Required

Differential distinction: Dysarthria (neuromuscular execution) vs Apraxia (motor planning) vs Aphasia (linguistic).

4. Measured Acoustic Biomarkers & Physical Signal Properties

Acoustic Biomarker	Measured Value	Reference Norms	Physiological Interpretation
Acoustic Speaker Gender	Male (88.5%)	Pitch F0 & Centroid	Vocal tract acoustic profile
Mean Pitch (F0)	147.9 Hz	M: 85-165Hz | F: 165-260Hz	Vocal fold vibrational frequency
Jitter (Local Perturbation)	2.67%	< 1.50% (Normal)	Pitch period vibration stability
Shimmer (Amplitude Perturbation)	18.20%	< 4.00% (Normal)	Glottal resistance & amplitude
Harmonics-to-Noise Ratio (HNR)	3.6 dB	> 15.0 dB (Normal)	Acoustic periodicity vs breathiness
Silence / Pause Dynamics	63.0%	10% - 25% (Typical speech)	Articulatory fluency & hesitations

5. Longitudinal Progress Comparison (Session-to-Session)

Metric	Previous Session	Current Session	Measurement Change
Intelligibility Index:	25.9%	44.1%	+18.2%
Speech Rate (WPM):	97.1 WPM	87.5 WPM	-9.6 WPM

Longitudinal Notice: Changes in acoustic measurements reflect session acoustic variability and do not prove clinical cure.

6. Clinical Next Steps & Professional Evaluation Plan

* Formal Motor Speech Evaluation by a licensed CCC-SLP (e.g., Frenchay Dysarthria Assessment-2 or Apraxia Battery for Adults-2).
* Comprehensive Language Assessment (e.g., Western Aphasia Battery-Revised or Boston Diagnostic Aphasia Examination) if word retrieval or syntax difficulties persist
* Laryngeal Videostroboscopy if dysphonia, vocal tremor, or vocal fold bowing is suspected.

SCREENING SUPPORT ONLY. REQUIRES FORMAL EVALUATION BY A CERTIFIED SPEECH-LANGUAGE PATHOLOGIST.

7. Clinician Review & Attestation Section

Review Status: PENDING FORMAL CLINICAL REVIEW **Reviewing Clinician: Not Assigned**

Clinician Notes: No clinician review notes recorded yet.

MANDATORY CLINICAL SCREENING & UNCERTAINTY DISCLAIMER:

SpeechCare AI is an assistive screening and clinical decision-support prototype. It does NOT provide a definitive medical diagnosis. All screening indicators, severity classifications, and acoustic indices account for potential limitations in audio quality, microphone hardware, and environmental noise. These findings must NEVER replace a comprehensive bedside evaluation by a certified Speech-Language Pathologist (CCC-SLP) or licensed neurologist. Further clinical assessment is strongly advised before therapeutic or pharmacological interventions.